

CarbaQ CO2 Capture Solution

Operations Manual

Version 2.0

Fernwick Labs

This manual describes installation, operation, maintenance and troubleshooting of the CarbaQ unit, which recovers CO2 from fermentation off-gas and stores it as a liquid in a receiver vessel.

Contents

1. Overview of the CarbaQ CO2 Capture Unit
2. Installation
3. Operating the Unit
4. Daily Checks
5. CarbaQ Software Components
6. Maintenance
7. Cleaning and Storage
8. Gas Quality Checks
9. Troubleshooting Tips
10. Service and Support

1. Overview of the CarbaQ CO2 Capture Unit

Fermentation off-gas enters the unit through the foam trap, passes the activated carbon bed that removes odours and volatile organic compounds, and is compressed, cooled in the chiller and condensed into the receiver vessel.

An oxygen sensor monitors the purity of the gas before it is condensed.

1.1 Process flow

Foam trap, activated carbon bed, compressor, chiller, receiver vessel. The heater keeps the gas above its dew point before the carbon bed.

1.2 Main components

The skid carries the compressor, the chiller, the carbon bed, the heater and the control cabinet with the HMI. The receiver vessel stands next to the skid.

1.3 Safety

CO₂ displaces oxygen. Operate the unit only in a ventilated area with a CO₂ monitor. The receiver vessel is a pressure vessel; never exceed its rated pressure.

1.4 Technical data

Capture rate, electrical supply and receiver capacity are given on the data plate of the unit.

2. Installation

Place the skid on a level floor near the fermentation vessels. Keep the inlet hose short and free of low points where foam can collect.

2.1 Electrical connection

Connect the unit to the supply given on the data plate. The unit must be earthed.

2.2 Preparing the receiver vessel

Attach the receiver vessel to the liquid outlet of the chiller and open the receiver vent to purge air from the line.

2.3 Setting the back pressure regulator

The back pressure regulator (BPR) keeps the receiver at the pressure needed for liquefaction. Set the BPR so that Receiver Pressure reads 140 psi with the unit running.

Turn the BPR adjusting screw in steps of about 1/8 of a turn and wait for the pressure to settle after each step. Lock the adjusting screw when the value is reached.

2.4 Leak check

Check all connections with leak detection spray before the first start.

3. Operating the Unit

Start the unit from the HMI. The compressor starts when the oxygen content of the gas is below the set limit. Stop the unit from the HMI at the end of the fermentation.

4. Daily Checks

Check the foam trap, the sight glass, the receiver pressure and the HMI alarm list every day.

5. CarbaQ Software Components

The HMI shows the process screen, the trend screen and the alarm screen. The alarm screen lists active alarms with their time stamp.

5.1 HMI Alarms

The unit raises alarms for high pressure, high temperature, low oxygen purity and compressor faults. Shutdown alarms stop the compressor and must be acknowledged before restart.

6. Maintenance

Follow the maintenance intervals below. Record every maintenance action in the service log.

6.1 Replacing Activated Carbon

Replace the activated carbon when the VOC alarm occurs or after the interval on the data plate. Stop the unit and depressurize the carbon bed before opening it.

6.2 Filters and drains

Replace the coalescing filter element and check the auto-drain valve every three months.

6.3 Calibration of Oxygen Sensor

Calibrate the oxygen sensor monthly with ambient air as span gas. Moisture on the sensor causes drift; dry the sample line before calibration.

The O2 Percentage value on the HMI must read the span value within tolerance after calibration.

7. Cleaning and Storage

Clean the foam trap and the inlet hose after every fermentation cycle.

7.1 Storage

For storage, depressurize the unit, drain the chiller and close all valves.

7.2 Seasonal shutdown

Before a seasonal shutdown replace the activated carbon and the filter elements.

8. Gas Quality Checks

Take a gas sample from the receiver and analyse oxygen, VOC and moisture. A foam trap that is not emptied regularly raises the VOC content.

9. Troubleshooting Tips

Problem	Possible cause	Remedy
Moisture in sight glass	Condensate carried over from the chiller	Drain the chiller separator and check the auto-drain valve.
Uneven gas flow; gas enters the unit in surges	Foam trap overfilled or a leak on the inlet hose	Empty the foam trap and tighten the inlet hose connections.
Chiller High Pressure Shutdown with moisture in the carbon bed	Moisture reached the activated carbon bed	Depressurize the unit, then dry or replace the carbon bed before restarting.
Receiver High Pressure Shutdown	Back pressure regulator set too high	Lower the BPR setting and vent the receiver slowly.
Chiller High Pressure Shutdown	Back pressure regulator was moved	Adjust the BPR about 1/8 of a turn at a time until Receiver Pressure reads 140 psi, then lock the adjusting screw. Allow up to 15 minutes for the pressure to settle after each step.
Compressor High Pressure Shutdown	Blockage downstream of the compressor	Check the auto-drain and the check valve for blockage.
Heater High Temperature Shutdown	Heater thermostat fault or no gas flow	Check gas flow and the heater thermostat.

9. Troubleshooting Tips (continued)

Problem	Possible cause	Remedy
Negative temperature but no liquid in the receiver	Receiver insulation damaged or vent open	Check the receiver insulation and close the vent valve.
Over pressurization of the compressor	Coalescing filter blocked or outlet valve closed	Replace the coalescing filter; open the outlet valve.
Compressor false operation	Contactors, breaker or VFD fault	Reset the breaker, check the contactors and the VFD fault log on the HMI.

10. Service and Support

For an urgent problem call the service hotline. For all other questions write to service@fernwick-labs.example and quote the serial number of the unit.

Revision History

Version 2.0: troubleshooting table extended; BPR procedure updated.